

Session 4: Inheritance Topics : Single, Multilevel, Hierarchical inheritance. Example: Content Creator ? YouTuber / Podcaster.

Tasks

- 1) Create a base class called **SocialMediaUser** with properties **username** and **followers**, and a method **displayProfile()** that prints the username and follower count.

Code-

```
#include<iostream>
using namespace std;
/*
1.
Create a base class called SocialMediaUser
with properties username and followers, and
a method displayProfile() that prints the
username and follower count.*/

class SocialMeadiaUser{

protected:

    string username;
    int followers;

public:
    SocialMeadiaUser(string u,int f)
    {
        username=u;
        followers=f;
    }

    void displayProfile(){
```

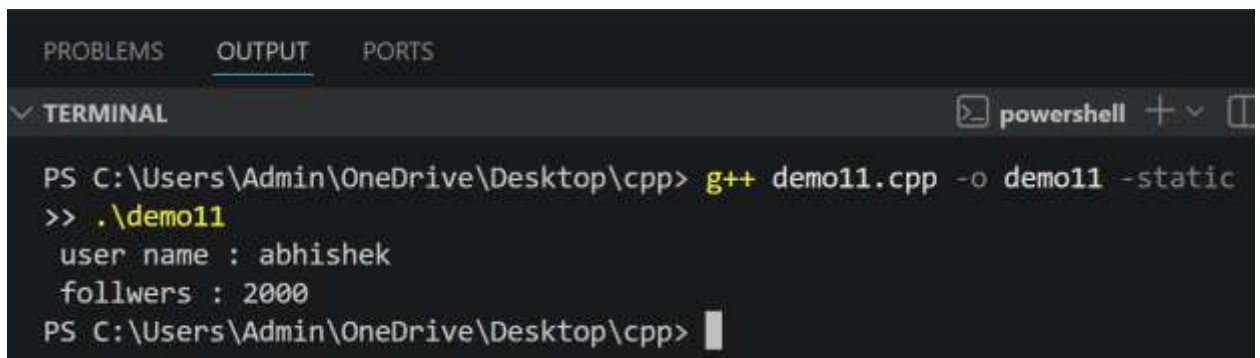
```
        cout<<" user name : "<<username<<endl;
        cout<<" follwers : "<<followers<<endl;

    }
};

int main(){
    SocialMeadiaUser s1("abhishek",2000);
    s1.displayProfile();
    return 0;

}
```

Output:



```
PROBLEMS  OUTPUT  PORTS
TERMINAL  powershell + v [ ]

PS C:\Users\Admin\OneDrive\Desktop\cpp> g++ demo11.cpp -o demo11 -static
>> .\demo11
    user name : abhishek
    follwers : 2000
PS C:\Users\Admin\OneDrive\Desktop\cpp> 
```

2) Build a YouTuber class that inherits from SocialMediaUser and adds a property channelName and a method uploadVideo(title) that prints 'Video [title] uploaded to [channelName]'

Code-

```
#include<iostream>
using namespace std;
/*
```

Build a YouTuber class that inherits from SocialMediaUser and adds a property channelName and a method uploadVideo(title) that prints 'Video [title] uploaded to [channelName]'.*/

```
class SocialMediaUser{

    protected:

    string username;
    int followers;

    public:
    SocialMediaUser(string u,int f)
    {
        username=u;
        followers=f;
    }

    void displayProfile(){

        cout<<" user name : "<<username<<endl;
        cout<<" followers : "<<followers<<endl;

    }
};

class Youtuber : public SocialMediaUser{
```

```

private:
string channelName;

public:
Youtuber(string u,int f,string c):SocialMeadiaUser(u, f)
{
    channelName=c;
}

void uplodatedVideo(string title){
    cout<<"Video ["<<title<<"] uploaded to ["<<channelName<<"]"<<endl;
}

};

int main(){

    Youtuber yt("abhishek",2000,"codewithabhi");

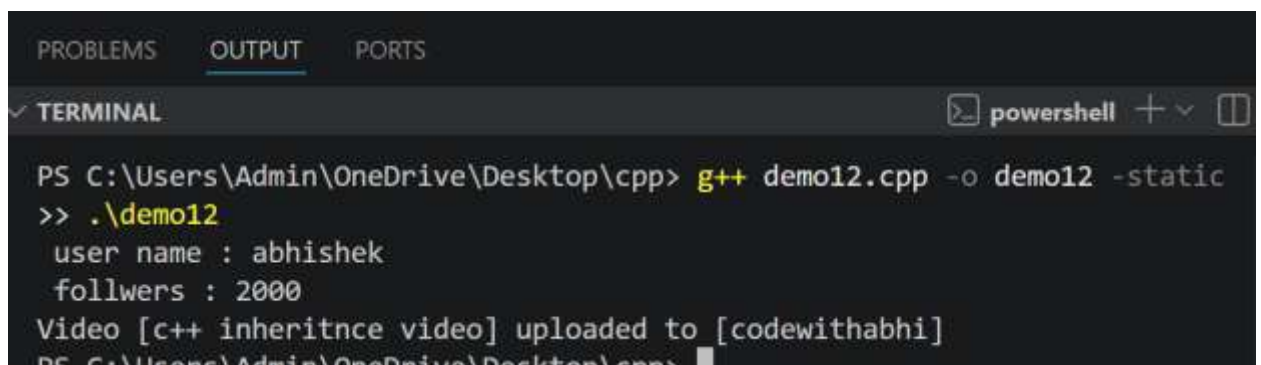
    yt.displayProfile();

    yt.uplodatedVideo("c++ inheritnce video");

    return 0;
}

```

Output:



```

PROBLEMS  OUTPUT  PORTS
✓ TERMINAL  powershell
PS C:\Users\Admin\OneDrive\Desktop\cpp> g++ demo12.cpp -o demo12 -static
>> .\demo12
    user name : abhishek
    followers : 2000
Video [c++ inheritnce video] uploaded to [codewithabhi]
PS C:\Users\Admin\OneDrive\Desktop\cpp>

```

3) Create a Podcaster class that also inherits from SocialMediaUser and adds a property podcastName and a method publishEpisode(episodeTitle) that prints 'Episode [episodeTitle] published on [podcastName]'.

Code-

```
#include<iostream>
using namespace std;

/*Create a Podcaster class that also inherits from SocialMediaUser
and adds a property podcastName and a method publishEpisode
(episodeTitle) that prints 'Episode [episodeTitle] published
on [podcastName]'*/

class SocialMediaUser{

protected:

string username;
int followers;

public:
SocialMediaUser(string u,int f)
{
username=u;
followers=f;
}

void displayProfile(){

cout<<" user name : "<<username<<endl;
cout<<" follwers : "<<followers<<endl;

}
};

class podcast : public SocialMediaUser{

private:
```

```

string podcastName;

public:
podcast(string u,int f,string p):SocialMeadiaUser(u, f)
{
    podcastName=p;
}

void publishEpisode(string Title){
    cout<<"Video ["<<Title<<"] uploaded to ["<<podcastName<<"]"<<endl;
}

};

int main(){

    podcast pt("abhishek",2000,"podcastwith Raj Shamani");

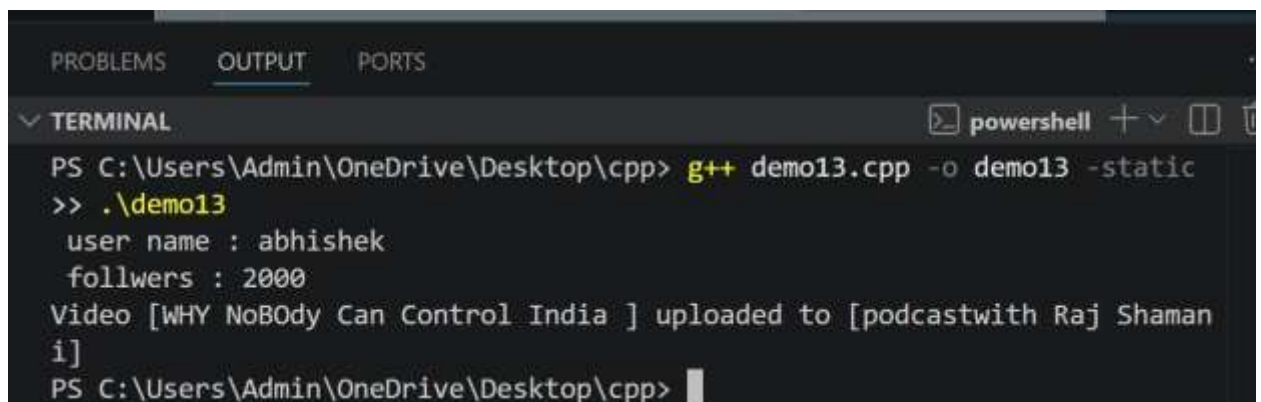
    pt.displayProfile();

    pt.publishEpisode("WHY NoBOdy Can Control India ");

    return 0;
}

```

Output:



```

PROBLEMS  OUTPUT  PORTS
TERMINAL
PS C:\Users\Admin\OneDrive\Desktop\cpp> g++ demo13.cpp -o demo13 -static
>> .\demo13
    user name : abhishek
    followers : 2000
Video [WHY NoBOdy Can Control India ] uploaded to [podcastwith Raj Shamani]
PS C:\Users\Admin\OneDrive\Desktop\cpp>

```

- 4) **Demonstrate multilevel inheritance by creating a class GamingYouTuber that inherits from YouTuber and adds a method streamGame(gameName) which prints '[username] is now streaming [gameName] on [channelName]'.**

Code-

```
#include<iostream>

using namespace std;

/*Demonstrate multilevel inheritance by creating a class
GamingYouTuber that inherits from YouTuber and adds
a method streamGame(gameName) which prints '[username] is now
streaming [gameName] on [channelName]'./

class SocialMediaUser{

    protected:

    string username;


    public:
    SocialMediaUser(string u)
    {
        username=u;

    }
```

```
void displayProfile(){

    cout<<" user name : "<<username<<endl;

}

};

class Youtuber : public SocialMeadiaUser{

protected:

    string channelName;

public:

    Youtuber(string u,string c):SocialMeadiaUser(u)
    {
        channelName=c;
    }

    void displayProfile2(){
        cout<<" channelName : "<<channelName<<endl;
    }

};
```



```
class GamingYoutber : public Youtuber{

    public:

    GamingYoutber(string u,string c):Youtuber(u,c)
    {

    }

    void method_streamGame(string gameName){

        cout<< "["<<username<<"] is now streaming ["<<gameName<<"] on
        ["<<channelName<<"]"<<endl;

    }

};

int main(){

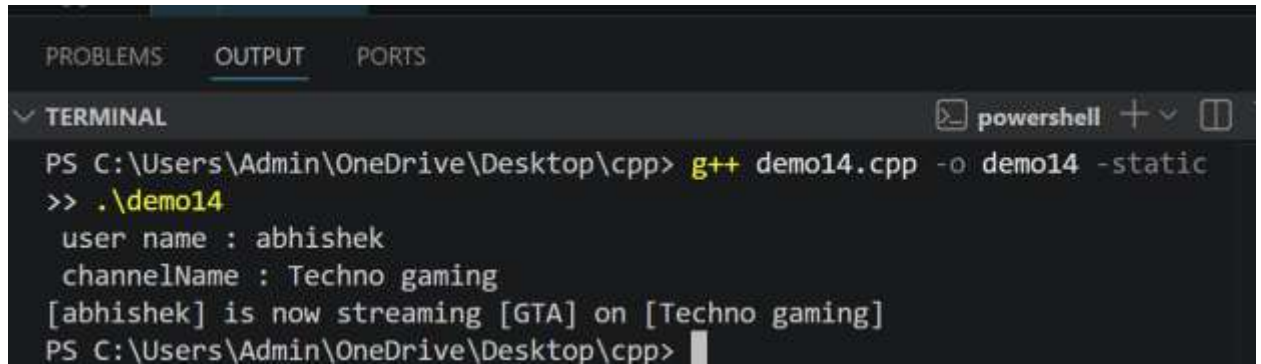
    GamingYoutber gt("abhishek","Techno gaming");

    gt.displayProfile();
    gt.displayProfile2();

    gt.method_streamGame("GTA");
```

```
return 0;  
}
```

Output:



```
PROBLEMS  OUTPUT  PORTS  
✓ TERMINAL  powershell + -  
PS C:\Users\Admin\OneDrive\Desktop\cpp> g++ demo14.cpp -o demo14 -static  
>> .\demo14  
user name : abhishek  
channelName : Techno gaming  
[abhishek] is now streaming [GTA] on [Techno gaming]  
PS C:\Users\Admin\OneDrive\Desktop\cpp>
```

5) Implement hierarchical inheritance by creating an InstagramInfluencer class that inherits from SocialMediaUser and adds a method postStory(storyTitle) which prints '[username] posted a new story: [storyTitle]'.

Hint: Think about how SocialMediaUser is the parent for YouTuber, Podcaster, and InstagramInfluencer.

Code-

```
#include<iostream>

using namespace std;

/*Implement hierarchical inheritance by creating an InstagramInfluencer class that
inherits from SocialMediaUser and adds a method
postStory(storyTitle) which prints '[username] posted a new
story: [storyTitle]'.  
  
Hint: Think about how
SocialMediaUser is the
parent for YouTuber, Podcaster, and InstagramInfluencer.*/

class SocialMediaUser{

protected:

string username;

public:

SocialMediaUser(string u)
{
```

```
username=u;

}

void displayProfile(){

    cout<<" user name : "<<username<<endl;

}

};

class InstagramInfluencer: public SocialMeadiaUser{

    public:
    InstagramInfluencer(string u):SocialMeadiaUser(u)
    {
    }

    void postStory(string storyTitle){
        cout<<"["<<username<<"] posted a new story: ["<<storyTitle<<"]"<<endl;
    }

};

int main(){
```

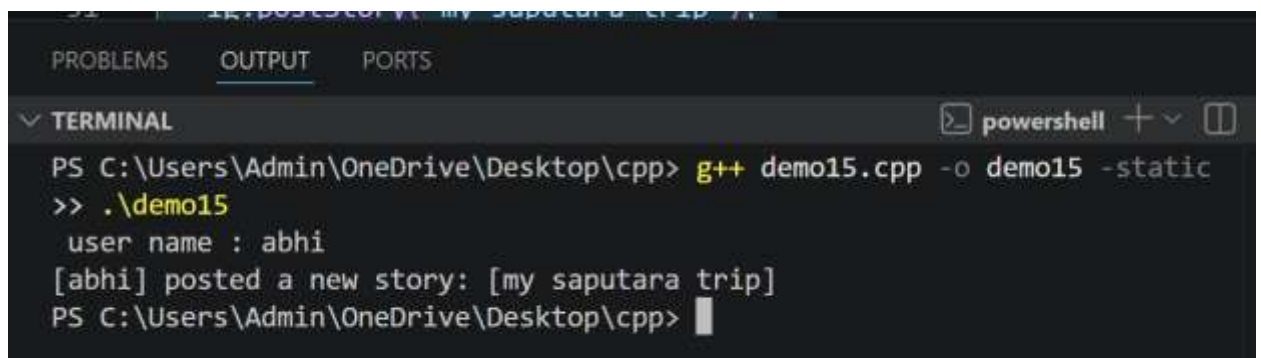
```
InstagramInfluencer ig("abhi");

ig.displayProfile();

ig.postStory("my saputara trip");

return 0;
}
```

Output:



```
PROBLEMS OUTPUT PORTS
▼ TERMINAL powershell + -
PS C:\Users\Admin\OneDrive\Desktop\cpp> g++ demo15.cpp -o demo15 -static
>> .\demo15
    user name : abhi
[abhi] posted a new story: [my saputara trip]
PS C:\Users\Admin\OneDrive\Desktop\cpp>
```